BRAC University — **Spring 2026: Midterm Examination**
Department of Computer Science and Engineering — **Set: B**
Course Code: CSE321 — **Duration: 1 Hour 30 Minutes**
Course Title: Operating Systems — **Total Marks: 25**

Name: ______________________________

ID: ______________________ **Section:** ______________

Instructions:

- Answer all questions.
- Marks are shown in brackets, and difficulty is shown as a tag (e.g., CA-E, CA-M, CA-D).
- Understanding the question is part of the exam.
- Write the answers for Questions 1(a), 1(b), 1(c), 2(a), 2(c), 2(d), and 3(a) in the designated answer boxes in this question paper.
- Write the answers for Questions 2(b), 3(b), and 3(c) in the answer script.
- For conceptual questions, write only the precise and relevant answer in short. Do not write detailed or descriptive explanations.

1. CO1 Process [8]

Consider the code snippet given below:

```c
void rainy() {
    int x = 5;
    printf("A ");

    if (fork() != 0) {

        x += 2;
        printf("B ");

        if (fork() == 0) {
            x *= 2;
            printf("C ");
        } else {
            wait();
            x -= 3;
            printf("D ");
        }

    } else {

        x += 4;
        printf("E ");
    }

    printf("F ");
}
```

a) Find out the total number of processes created by this function (including the original process that called rainy)? **CA-M [2]**

Answer:

b) **Identify** which of the following outputs are possible from the code given above? **(Select all possible outputs)**. **CA-M [3]**

(i) A B D F C F E F
(ii) A B C F D F E F
(iii) A E F B C F D F
(iv) A B E C F F F D
(v) A B C F E F D F

Answer:

c) Suppose the **line 25** of the code given above has been changed from **printf("F ");** to **printf("%d",x);**. **Identify** all possible values of **x** that could be printed out after the modification. **(Select all possible outputs)**. **CA-D [3]**

(i) 9
(ii) 0
(iii) 14
(iv) 4
(v) 13

Answer:

2. CO1 CPU Scheduling **[11]**

a) In an alternate implementation of a multi-level feedback queue, the priority of each process is stored in its PCB (Process Control Block), and a single queue is used instead of having multiple logical queues. All other aspects of the MLFQ remain unchanged.
Explain whether there is any difference in terms of scheduling performance if this alternate version is implemented. **CA-D [2]**

Answer:

b) In an OS, **Multilevel Feedback Queue** functions as a scheduler. In the scheduler, the ready queue is divided into **three** levels. At every level, the round-robin algorithm has been employed, and the following constraints must be maintained.

- **Q1**: time quantum = **3**, priority = **0**
- **Q2**: time quantum = **5**, priority = **1**
- **Q3**: time quantum = **8**, priority = **2**

Here, a lower priority value indicates higher priority; upon completion of an I/O operation, the scheduler will promote a process by one level. The information on burst times, arrival times, and I/O times of three processes is given below: **CA-E** [**5**]

Process	**Burst Time**	**Arrival Time**	**I/O Time**
P3	8	0	5s I/O [after total 5s CPU allocation]
P2	11	1	N/A
P1	8	2	N/A

(i) Draw a Gantt chart using the multilevel feedback queue scheduling algorithm, showing the states of the ready queues of different levels. [**4**]

(ii) Calculate the average turnaround time from the Gantt chart. [**1**]

c) A system uses a preemptive scheduling algorithm (SRTF). A process is executing, and a new process arrives with a shorter CPU burst. **CA-E** [**2**]

(i) Explain what happens in this situation in terms of average response time and justify whether this behavior improves system performance based on the mentioned parameters. [**1**]

(ii) Mention one possible drawback. [**1**]

Answer:

I	
II	

d) A university computer lab runs two different operating systems:

- **System A** is used in a batch-processing server where jobs are submitted and executed using **FCFS scheduling**.
- **System B** is used in a student interactive terminal system where multiple users run programs simultaneously using **Round Robin with a small time quantum**.

Analyze, and **write** the behavior of both systems in terms of Response Time and Context Switch Overhead. **CA-M [2]**

Answer:

3. CO2 Threads & Process Synchronization [6]

a) A multi-threaded banking system updates account balances. A thread may be cancelled while updating shared data. Two approaches are used:

- **Design A:** Asynchronous cancellation
- **Design B:** Deferred cancellation

Explain which design is better and why? **CA-M [2]**

Answer:

b) A program uses 4 threads to process large amounts of data. Each thread processes a portion of the data on a 4-core system. **CA-E [2]**

(i) Identify which benefit of multithreading is demonstrated in this scenario. [1]

(ii) Explain if a single-threaded program would perform better or worse in this situation. [1]

c) Consider the following implementation of *compare_and_swap*(CAS) and a function *decrement* that uses CAS:

```
int CAS(int *v, int expected, int new) {
    int temp = *v;
    if (*v == expected)
        *v = new;

    return temp;
}
```

```
void decrement(int *i) {
    int old = *i;
    while (CAS(i, old, old - 1) != old) {
        old = *i;
    }
    print(*i);
}
```

Suppose the initial value of i is **10**, and **10** threads concurrently call the *decrement* function, passing $\&i$ as the argument. **Write** the output of the program and **justify** it. **CA-D [2]**